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2026 CASE COMPETITION

Brand New Market: AI Liability Insurance

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Section 1: Industry Background

You are a strategy advisor for **TileShield Insurance (TSI)**, a mid-sized Canadian property & casualty insurer headquartered in Toronto. Since 1998, TSI has built a stable business insuring small and medium-sized companies. It operates across five lines: Commercial General Liability and Commercial Auto, plus three specialty lines built for fast-moving risks: Errors & Omissions, Cyber Liability, and Technology Products Liability. Many of TSI's clients work in technology, software, online services, and professional services.

That client base puts TSI in the middle of the biggest shift in business risk in a generation. The board has retained your team to answer one question: should TileShield enter the new market for **AI liability insurance**? And if so, how? To advise them, you first need to understand the world TSI is stepping into.

AI is everywhere — and so are its mistakes

In just a few years, artificial intelligence has moved from research labs into everyday business. Law firms use it to draft legal filings, hospitals use it to read scans, banks use it to score credit and spot fraud, and retailers, airlines, and even governments have wired it into the systems their customers rely on. The value created is enormous, but so is a brand-new kind of risk. AI can confidently produce answers that are completely made up, reproduce human bias at a massive scale, leak private data, or malfunction in the physical world.

How AI fails — and what it costs

Real-world AI failures tend to fall into a handful of repeating patterns. Each one creates a different kind of legal and financial exposure for the businesses that deploy these tools:

- **Hallucinated output.** AI invents facts and states them confidently. A New York law firm was sanctioned in 2023 after filing a brief that cited six court cases ChatGPT had fabricated; an airline was forced to honour a refund policy its chatbot made up.
- **Algorithmic bias.** AI systematically disadvantages certain groups. An online tutoring company paid $365,000 to settle with the U.S. Equal Employment Opportunity Commission after its hiring software was set to automatically reject older applicants (EEOC v. iTutorGroup, 2023).
- **Privacy and security failures.** AI tools mishandle sensitive data. For example, hospitals route patient records through outside AI services, triggering privacy-law violations (Abridge AI Lawsuit).
- **Cyber exploitation via AI.** Attackers use AI to break in or scale up fraud, making cyberattacks faster, cheaper, and harder to spot (an AI-enabled cyberattack reported by Anthropic).
- **Misinformation and deepfakes.** AI convincingly impersonates real people. In 2024, a Hong Kong company lost $25 million after an employee was tricked into wiring funds during a deepfake video call faking the CFO.

- **Autonomous system failure.** Physical AI breaks down in the real world. In 2023, Cruise robotaxis repeatedly stalled and blocked San Francisco streets, and the company's operating permits were suspended after a pedestrian was injured (SF Standard); factory robots have also caused fatal workplace accidents (CBS News).

These are not rare one-offs. Public trackers such as the AI Incident Database now log thousands of cases, and U.S. lawsuits tied to generative AI have **grown more than tenfold since 2020, passing 700 cases by 2025**. Outcomes range from quiet settlements to multi-million-dollar class actions and regulatory fines.

The insurance industry is split

Insurers have watched these losses pile up and reacted in three very different ways:

- **Pulling back.** Many traditional carriers are adding blanket "AI exclusions" to their policies, meaning if a loss involves AI at all, it isn't covered. Standardized exclusion wording is already being adopted across North America, protecting insurers but leaving businesses with a widening coverage gap.
- **Leaning in.** A small group of specialists (names like Armilla and Testudo) has launched dedicated AI insurance products aimed at specific failures such as poor model performance, hallucinations, and bias. The market is tiny but growing fast: one widely cited forecast projects it could reach roughly $4.7 billion by 2032.
- **Watching and waiting.** Most insurers are doing neither, monitoring losses and building expertise while they wait for clearer data and rules. The danger of waiting is being shut out of a market that takes shape without them.

The rules are still being written

Regulators are moving too, sometimes faster than insurers. The European Union's AI Act now sorts AI systems by risk level and places duties on both the builders and users of AI. Canada has proposed similar rules for "high-impact" systems, and U.S. agencies have already taken action against companies whose AI harmed consumers. For insurers, this cuts two ways: new rules create new ways to be sued (and therefore new claims), but they also begin to clarify who is responsible when AI fails.

Section 2: Company Overview — TileShield Insurance

Company Snapshot

Company	TileShield Insurance (TSI)
Founded	1998, Toronto, Ontario
Type	Mid-sized Canadian Property & Casualty (P&C) insurer
Client Focus	SMEs (Small and Medium-sized Enterprises) across the commercial, professional, transportation, and technology sectors
Approximate Canadian P&C Market Share	~1%, based on earned premium/insurance revenue scale
2025 Earned Premium	$834.2M across current lines
Current Lines of Business	Commercial General Liability · Commercial Auto Liability · E&O · Cyber Liability · Technology Products Liability

Since its founding in 1998, the company has established its presence through traditional commercial insurance businesses, particularly Commercial General Liability and Commercial Auto Liability. After building strong roots in the insurance industry, TSI gradually expanded into specialty commercial lines and began primarily serving small-to-medium enterprises (SMEs), particularly in technology, online services, software, and professional sectors. Over time, this expansion led the company to develop several specialty liability businesses, most notably Errors & Omissions (E&O), Cyber Liability, and Technology Products Liability.

Current Situation

TileShield Insurance (TSI) is currently at a strategic inflection point. Over the past two years, the company's board of directors has become increasingly concerned about the sustainability of its core business lines. While TSI has historically been profitable, though more volatile, in cyber experience and E&O (Errors & Omissions) insurance, shifts in the technology landscape have begun to affect both client demand and the nature of risk exposures. Several long-standing clients in digital services (web, design, SEO), online education, and content-based industries

(copywriting, stock media) have experienced disruption from AI-driven solutions, leading to slower growth and, in some cases, reduced demand for traditional coverage.

At the same time, management has observed that many of these same clients are rapidly adopting artificial intelligence in their operations, introducing new types of risks that are not fully addressed by existing products. In recent board discussions, executives have raised concerns that the company may be underexposed to emerging growth opportunities, particularly as competitors begin to explore AI-related liability offerings. This has created pressure for the board to make a timely decision on whether TSI should enter this market.

However, the challenges associated with this decision are significant. Unlike traditional lines of business, AI liability comes with little to no credible historical loss data, making it difficult to estimate expected losses with confidence or establish reliable pricing benchmarks. In addition, the legal environment surrounding artificial intelligence remains uncertain, with ongoing debate over whether responsibility should rest with developers, users, model providers, or other third parties when AI-related failures occur. These uncertainties complicate both underwriting and product design, particularly in a market where claim patterns and liability standards are still evolving.

Despite these challenges, TSI is not entering the discussion without relevant experience. Through its Errors & Omissions, Cyber Liability, and Technology Products Liability businesses, the company has developed underwriting expertise in technology-related risks and has built long-standing relationships with small-to-medium enterprises operating in sectors where AI adoption is accelerating most rapidly. Management has experience adapting policy structures and underwriting approaches to emerging risks, providing a foundation from which AI liability coverage could be developed. Nevertheless, the board recognizes that AI liability introduces exposures that are not fully comparable to any existing line of business, and that the absence of a dedicated underwriting framework or meaningful claims history increases the risk of pricing inadequacy. As a result, the central question facing the board is no longer whether AI liability represents a potential opportunity, but whether TSI can enter the market in a disciplined manner that balances growth ambitions with prudent risk management.

Section 3: TileShield Financial Performance

TileShield's five-year results by line of business (earned premium, claim counts, incurred losses, expenses, and earned exposure) are provided in Exhibit A. Teams should use this data to assess the company's underwriting performance and its readiness to take on a new risk (see Assignment 1).

TSI Financial Data

Errors & Omissions

Year	2021	2022	2023	2024	2025
Exposure	4,150.00	4,413.00	4,592.00	4,889.00	5,012.00
Claim_Frequency	0.1222	0.1147	0.122	0.1168	0.1201
Earned_Premium	78924.00	70402.90	93844.40	88885.00	105606.10
Claim_Count	507.00	506.00	560.00	571.00	602.00
Average_Severity	91.00	89.80	103.80	99.50	99.60
Operating_Expenses	19731.00	17600.70	23461.10	22221.20	26401.50

Cyber Liability

Year	2021	2022	2023	2024	2025
Exposure	1,826.00	2,110.00	2,788.00	3,375.00	3,998.00
Claim_Frequency	0.2371	0.0981	0.1438	0.1446	0.0768
Earned_Premium	122280.20	68400.80	138581.40	138679.10	132066.50
Claim_Count	433.00	207.00	401.00	488.00	307.00
Average_Severity	164.40	208.60	222.80	195.60	299.60
Operating_Expenses	34238.40	19152.20	38802.80	38830.10	36978.60

Technology Products Liability

Year	2021	2022	2023	2024	2025
Exposure	953.00	1,002.00	1,075.00	1,282.00	1,430.00
Claim_Frequency	0.1595	0.1936	0.1553	0.1810	0.1664
Earned_Premium	51594.80	46256.70	54527.30	101136.20	100729.70
Claim_Count	152.00	194.00	167.00	232.00	238.00
Average_Severity	237.30	164.90	199.30	226.90	231.80
Operating_Expenses	12898.70	11564.20	13631.80	25284.00	25182.40

Commercial General Liability

Year	2021	2022	2023	2024	2025
Exposure	12,162.00	12,199.00	11,779.00	12,041.00	12,450.00
Claim_Frequency	0.1190	0.1176	0.1234	0.1237	0.1178
Earned_Premium	212676.10	210739.90	212113.20	217254.90	186977.80
Claim_Count	1447.00	1435.00	1453.00	1490.00	1467.00
Average_Severity	80.10	81.20	81.10	80.00	70.70
Operating_Expenses	46788.70	46362.80	46664.90	47796.10	41135.10

Commercial Auto Liability

Year	2021	2022	2023	2024	2025
Exposure	8,946.00	9,247.00	9,132.00	9,369.00	9,454.00
Claim_Frequency	0.1076	0.1112	0.0922	0.1654	0.1477
Earned_Premium	140,958.10	186,716.20	124,072.10	297,441.50	308,841.70
Claim_Count	963.00	1,028.00	842.00	1,550.00	1,396.00
Average_Severity	86.50	108.80	89.60	115.50	131.10
Operating_Expenses	35,239.50	46,679.00	31,018.00	74,360.40	77,210.40

Section 4: The Three Strategic Options

The board has narrowed its decision to three options. Participants can evaluate all three and recommend one, or propose a justified hybrid or alternative.

Option A — Launch a Standalone AI Liability Product

TileShield could launch a new insurance product covering AI failures. This approach would position TSI as an early mover in a developing market and allow the company to build expertise and market presence ahead of competitors.

Option B — Add AI-Specific Endorsements to Existing Policies

TileShield could introduce AI-specific endorsements within its existing commercial liability policies. This would primarily target current commercial clients that are beginning to use AI and want coverage for AI-related exposures arising from their operations.

Option C — Build Capability before Market Entry

TileShield could hold off on offering AI liability coverage immediately while investing in the capabilities needed for future market entry. This would focus on preparing the company rather than launching coverage now.

Section 5: Assignment

Your team has been retained as outside advisors to the board of TileShield Insurance. The board is preparing to decide whether TileShield should enter the AI liability insurance market and, if so, how. Your final deliverable is a presentation to the board that addresses the three questions below.

You may use any of the data in Exhibits A through C, and are encouraged to bring in external research where it strengthens your case. Your recommendation should be supported by both quantitative analysis and clear strategic reasoning.

1. **Assess TileShield.** Using Exhibit A, evaluate TileShield's current financial position across its specialty lines (E&O, Cyber Liability, Technology Products Liability) and traditional lines (Commercial General Liability, Commercial Auto Liability). Judge what its recent financial experience says about its readiness to take on a new, emerging risk.

2. **Analyse the AI liability risk.** Using the AI incident dataset in Exhibit B, analyse whether AI liability is currently insurable for a mid-sized specialty insurer like TileShield. It is up to you to decide which patterns in the data to examine and which factors matter most to that judgment. Consider how the dataset can support your argument (Note that the AI incident dataset is based on external data, not TSI's own data).
3. **Make your recommendation.** Evaluate the three strategic options set out in Section 4 or propose a justified alternative. Tell the board what you would do and why. If you choose an alternative approach, please explain why you chose that over the three options provided. Explain the reasoning behind your recommendation, including the main risks it carries and how you would manage them.

Section 6: Submission Instructions

Submit your presentation as a PDF named: TeamName.pdf

Submission deadline: August 9th at 11:59 PM EST

Submit via: https://forms.gle/zeBTj1kYkJhhF4m1A

Case questions: quantify.case@gmail.com

General questions: uwquantify@gmail.com

Section 7: Glossary & Further Readings

Insurance terms

Term	Definition
Earned Premium	The portion of an insurance premium that corresponds to coverage that has already been provided. If a one-year policy is six months in, half of the premium has been earned.
Incurred Losses	The total cost of claims an insurer expects to pay for a given period, including both claims that have been settled and reserves set aside for claims that have been reported but not yet paid.
Loss Ratio	Incurred losses divided by earned premiums. A core measure of underwriting profitability (**the lower, the better**). A loss ratio above 100% means an insurer is paying out more in claims than it collects in premium.
Expense Ratio	Operating expenses divided by earned premiums. Captures the cost of running the business (underwriting, marketing, administration).
Combined Ratio	Loss ratio plus expense ratio. The single most important measure of

Term	Definition
	underwriting performance. Below 100% = underwriting **profit**. Above 100% = underwriting loss (which may still be offset by investment income).
Long-tail Line	An insurance line where claims often take years to be reported and settled (e.g., commercial general liability, workers' compensation). Long-tail lines require adequate reserves and capital to support claims that may emerge or settle over many years.
Commercial General Liability (CGL)	A core commercial line covering a business against third-party claims for bodily injury, property damage, and personal injury (such as libel or slander) arising from its ordinary operations, products, or premises. One of the most common forms of business liability coverage.
Commercial Auto Liability	Insurance covering a business against liability for bodily injury and property damage caused by vehicles owned or operated for business purposes (the company's legal responsibility when a business vehicle is involved in an accident).
Errors & Omissions (E&O)	Insurance that protects a company against claims that it provided inadequate professional services or made a professional mistake that caused financial harm to a client. Sometimes called "professional liability."
Cyber Liability	Cyber Liability covers losses from data breaches, ransomware, hacking, security failures, and certain network and business interruption events, depending on policy wording.
Technology Products Liability	Insurance covering loss caused by the failure of a technology product, whether hardware or software, including bodily injury, property damage, and financial loss suffered by customers or third parties as a result of the malfunction.
Endorsement (or Rider)	An add-on to an existing insurance policy that modifies the coverage by either adding protection for a specific risk or excluding it. Examples include add-ons covering claims from inaccurate or biased AI outputs, AI-driven security failures, and deepfake-enabled fraud.
Exclusion	A provision in an insurance policy that explicitly states certain risks are not covered. Many traditional insurers are now adding AI-related exclusions to standard policies.
Tail Risk	In insurance, the "tail" is the far end of a loss distribution: the rare, extreme outcomes where the largest claims sit. Tail risk is the danger that one of these low-probability, high-severity events produces losses far larger than a normal year, potentially

Term	Definition
	dominating the portfolio's result.
Frequency-Severity Modeling	An actuarial approach that models how often losses occur (frequency) and how large they are when they do occur (severity), separately. Multiplying the two gives an estimate of expected loss.

AI terms

Term	Definition
AI Hallucination	When an AI system generates output that is factually incorrect or fabricated, while presenting it confidently as fact. (including fake citations, invented policies, or non-existent events)
Algorithmic Bias	Systematic errors in AI outputs that disadvantage certain groups, typically arising from biased training data or flawed model design. Can result in discrimination claims under employment, lending, or civil rights laws.
Deployer	The company or organization that uses an AI system, as opposed to the developer who built it. Most AI liability claims to date have targeted deployers, since they are the party that put the AI in front of the customer or end-user.
Foundation Model	A large, general-purpose AI model (such as GPT-4, Claude, or Gemini) trained on broad data and adapted for many downstream uses. Foundation models are typically built by a small number of large AI developers and licensed to deployers.
Third-party AI vs. In-house AI	Third-party AI is licensed from an external provider (e.g., a deployer integrating ChatGPT into its workflow). In-house AI is built and trained by the company itself. Liability and data exposure differ between the two.
Deepfake / Synthetic Media	AI-generated audio, video, or images that convincingly impersonate real people or events. Used in fraud, impersonation scams, and reputation attacks.
EU AI Act	European Union legislation, in force from 2024, that classifies AI systems by risk level (unacceptable, high, limited, minimal) and imposes corresponding obligations on developers and deployers.

Further reading

Participants are encouraged but not required to consult these public resources for context.

1. **AI Incident Database (AIID) —** incidentdatabase.ai. The most comprehensive public catalog of AI failures, covering thousands of incidents across industries.

2. **AI Hallucination Cases Database —** www.damiencharlotin.com/hallucinations. Tracks legal cases in which AI systems have fabricated citations or facts in court filings.
3. **GWU AI Litigation Database —** blogs.gwu.edu/law-eti/ai-litigation-database. Tracks U.S. AI-related lawsuits, including who is being sued, on what theory, and how cases are being resolved.
4. **EU AI Act —** artificialintelligenceact.eu. Plain-language summary of the European AI regulation framework.